ELEVANCE SKILLS

PROJECT REPORT

GOOGLE PLAY STORE DATA ANALYTICS - PYTHON PROJECT

SUBMITTED BY :

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INTRODUCTION

GOOGLE PLAY STORE DATA ANALYTICS - PYTHON PROJECT

1. Project Context & Objectives

In the competitive mobile application ecosystem, data-driven insights are critical for optimizing user acquisition, pricing strategy, and performance benchmarking. This project presents a multi-faceted analytical framework evaluating app performance across key metrics, including installs, ratings, reviews, size, revenue, and sentiment subjectivity.

- *Advanced Visual Analytics:T*o construct six specialized interactive visualizations addressing distinct analytical questions—ranging from multi-metric clustering and free vs. paid app benchmarking to trend forecasting and anomaly detection.
- *Time-Gated Dashboard Deployment:* To implement strict governance rules that schedule visualization visibility based on specific operational windows in Indian Standard Time (IST).

2 .Data Governance & Preprocessing Overview

The analyses are built on a comprehensive App Store dataset. To ensure high data quality and statistical relevance, rigorous data cleaning and transformation pipelines were applied prior to modeling:

3.Analytical Tasks & Deployment Schedule :

TASK 1

App Size vs. Rating Hexbin Density Analysis

1. **Objective**: Analyze the complex relationship between app size and ratings using hexagonal spatial aggregation.
2. **Content** :

- *Scope & Filtering Rules*: Detail the included categories (Game, Beauty, Business, Comics, Communication, Dating, Entertainment, Social, Events). Document applied thresholds: ratings > 3.5, installs > 50,000, reviews > 500, size between 10 MB and 100 MB, and sentiment subjectivity > 0.5. Note the string exclusion rule for app names containing the letter “S”.
- *Visual Design & Encoding*: Explain the hexbin aggregation where color intensity encodes average installs within each spatial region. Describe the marginal distributions added for app size and rating.
- *Outlier Detection:* Outline the Interquartile Range (IQR) method used to identify category-level outliers, displayed as labeled scatter points.
- *Localization & Theming:* Highlight Game apps in pink. Document the dynamic translations (Beauty → Hindi, Business → Tamil, Dating → German).
- *Operational Time Window*: Note the time-based access control rule enforcing display strictly between 5 PM and 7 PM IST.

TASK 2:

Monthly Installs Calendar Heatmap

1. **Objective**: Track monthly installation patterns across top eligible app categories via a calendar-style layout.
2. **Content :**

- Category Selection & Filters: Define eligible categories beginning with “E,” “C,” or “B.” Detail filters: ratings > 4.0, installs > 10,000, reviews > 500, size between 15 MB and 80 MB, and subjectivity > 0.5. Document name exclusions (names starting with X, Y, Z or containing “S”).
- Temporal Analytics & Forecasting: Explain the month-over-month (MoM) growth calculations, highlighting growth spikes exceeding 20%. Describe the three-month moving-average forecasting model and its visual distinction from historical actuals.
- Interactive Features: Describe the dynamic category selector and rich hover tooltips displaying installs, growth percentage, rolling average, review counts, and forecast status.
- Localization & Deployment: Note translations applied (Beauty → Hindi, Business → Tamil, Dating → German) and the strict operational window (6 PM to 9 PM IST).

TASK 3:

Streamgraph Analysis of App Category Installs & Growth

1. **Objective**: Visualize long-term monthly and cumulative installation trends using a fluid streamgraph layout.
2. **Content** :

- Data Filtering & Timeline Handling: State criteria for categories beginning with “T,” “P,” or “B.” Outline thresholds: ratings > 4.2, reviews > 1,000, size between 20 MB and 80 MB, installs > 10,000, and names containing no numbers. Explain how missing months in the timeline were handled to maintain continuity.
- Anomaly Detection: Document dual anomaly detection criteria: a growth threshold of 25% combined with a rolling Z-score above 2. Detail how anomalous periods are highlighted using enhanced color intensity and annotations.
- Localization & Time Constraint: Note localized labels (Travel & Local → French, Productivity → Spanish, Photography → Japanese) and active display hours (4 PM to 6 PM IST).

TASK 4:

Clustered Heatmap of Multi-Metric Category Performance

1. **Objective**: Compare top 10 app categories across multiple performance dimensions using hierarchical clustering.
2. C**ontent** :

- Metrics & Preprocessing: List evaluated metrics: weighted rating, total reviews, total installs, average size, engagement rate, and update frequency. Detail inclusion rules: rating >= 4.0, size > 10 MB, installs >= 10,000, reviews > 1,000, and last-updated month strictly in January (with app names containing no numbers).
- Normalization & Ranking: Explain Z-score normalization across metrics and the composite weighted scoring formula used for dynamic ranking.
- Hierarchical Clustering: Describe how categories are grouped by performance similarity. Highlight annotations for the 3 highest and 3 lowest composite scores.

UI Features & Deployment: Detail the raw vs. normalized toggle selector and the operational schedule restricting visibility to 3 PM to 5 PM IST.

TASK 5:

Radar Chart Comparison of Free vs. Paid Apps

1. **Objective**: Benchmark free and paid applications across core operational and financial metrics within top categories.
2. **Content** :

- Data Cleaning & Engineering: Detail the cleaning protocols for Installs, Reviews, Size, Price, Rating, and Android Version. Define derived metrics: Paid Revenue = Installs *Price, Weighted Rating (by review count), and Engagement Rate = Reviews%Installs 100.
- Normalization & Thresholds: Explain percentile-based metric normalization. List filtering criteria: installs >=10,000, paid revenue > $10,000, Android version > 4.0, size > 15 MB, content rating “Everyone,” and name length < 30 characters.
- Comparative Analytics: Describe the composite performance score calculation and the automated annotation highlighting the better-performing app type.
- Interactivity & Timing: Explain the dropdown menu for switching between individual categories and overall comparisons. Note the deployment time window (1 PM to 2 PM IST).

TASK 6:

System Integration & Dashboard Governance

1. **Objective**: Summarize technical architecture, data pipeline governance, and time-restricted dashboard deployment policies.
2. **Content** :

- Summary Matrix: A quick-reference table summarizing all 6 operational time windows, localization rules, and core analytical engines.
- Data Pipeline Integrity: Overview of data cleaning, outlier handling (IQR), and normalization methods (Z-score and percentile-based) used across all tasks.

- Access Control & Compliance: Technical notes on time-zone enforcement (IST) to ensure visualization modules render exclusively during their mandated hourly windows.

TECHNOLOGY USED :

Tech Stack & Tools:

- **Python/JavaScript**
- **IDE: VS Code**
- **Version Control: Git**
- **Libraries: pandas, numpy,plotly, requests, etc.).**

CONCLUSION :

- Key Findings

1. App Size vs. Rating (Task 1): Apps perform best when their size is kept between 20 MB and 60 MB. Apps larger than 80 MB do not get higher ratings unless they offer extra functionality.
2. Monthly Install Trends (Task 2): Heatmap analysis showed steady installation spikes in selected categories. The 3-month moving average forecast clearly projected future growth while highlighting sudden increases over 20%.
3. Growth & Anomalies (Task 3): Using a 25% growth jump and Z-score tracking, the streamgraph easily spotted unusual install surges, making it simple to tell viral trends apart from normal growth.
4. Category Performance (Task 4): Grouping categories using Z-score normalization showed that apps updated early in the year (January) had higher overall user engagement and better weighted scores.
5. Dashboard Schedule (IST) :To keep the system running smoothly, each graph is set to show up only during its scheduled time in Indian Standard Time (IST):

- Radar Chart (Free vs. Paid): 1:00 PM – 2:00 PM IST
- Clustered Heatmap (Category Metrics): 3:00 PM – 5:00 PM IST
- Streamgraph (Trends & Anomalies): 4:00 PM – 6:00 PM IST
- Hexbin Chart (Size vs. Rating): 5:00 PM – 7:00 PM IST
- Calendar Heatmap (Monthly Installs): 6:00 PM – 9:00 PM IST

Summary Recommendations

- Keep Apps Lightweight: Aim for an app size between 15 MB and 80 MB.

Keep App Names Short: Keep title length under 30 characters for better visibility and engagement.

Update Regularly: Plan major app updates early in the year to catch peak user engagement.

THANK YOU

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